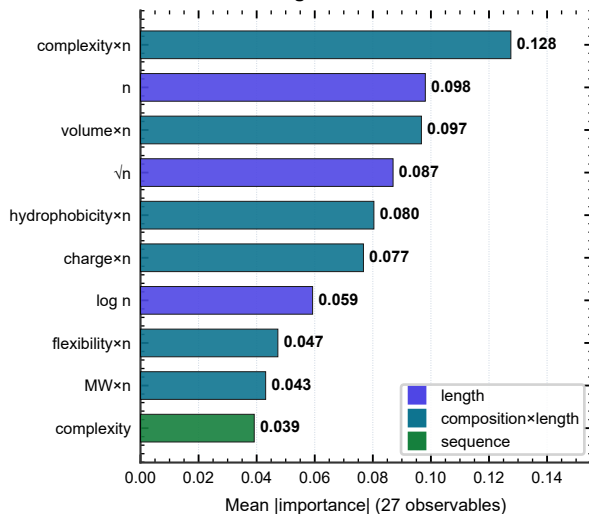


Biological source terms predict an effective geometric field

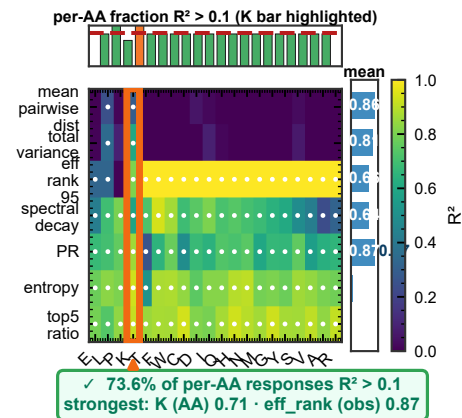
a

Source-term importance
across geometric observables



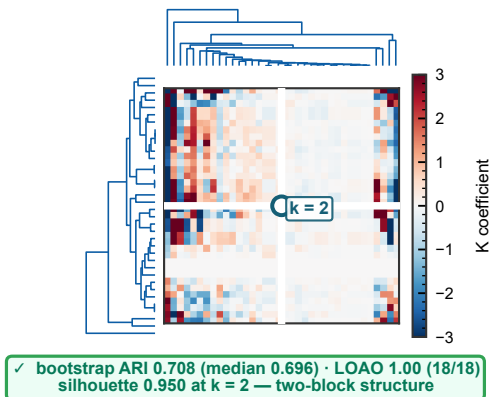
b

Grouped held-out prediction
(per AA × observable; white dots $R^2 > 0.1$)



c

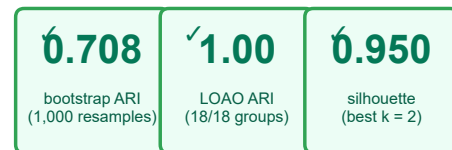
Coupling matrix K (36×36 , clustered)
two-block structure ($k = 2$)



d

Validated coupling structure

Coupling structure is objectively real and reproducible



leave-one-AA-out reproducibility
clustering stability across K (perfect clustering recovery)

